

The fundamental inequalities

Proposition (QR decomposition)

$A \in M_{n \times m}(\mathbb{R})$ with linearly independent columns

Then we can write

$$A = QR$$

$n \times m$ $m \times m$

where

- Q has orthonormal columns
- R is upper triangular & invertible

Rmk. This is simply a reformulation of Gram-Schmidt.

Observation (1) Q & R are uniquely determined (up to signs)

Pf. $A = QR$

$\parallel$

$$Q'R'$$

$$\Rightarrow Q'R' = QR$$

$$\Rightarrow Q'R'R^{-1} = Q$$

$\cdot R^{-1}$ $\underbrace{\quad}_{\text{upper}}$

orthogonal
columns

$$[u'_1 \ u'_2 \ \dots \ u'_m] \begin{bmatrix} a_{11} & 0 & \dots & 0 \\ 0 & a_{22} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & a_{mm} \end{bmatrix} = [u_1 \ u_2 \ \dots \ u_m]$$

Q' Q

$R' R^{-1}$

$$|a_{11}| \cdot \|u'_1\| = \|a_{11} u'_1\| = \|u_1\| = 1 = \|u'_1\|$$

$$\Rightarrow |a_{11}| = 1 \Rightarrow a_{11} = \pm 1$$

$$|a_{21}| \|u'_1\| = \|a_{21} u'_1 + a_{22} u'_2\| = \|u_2\| = 1$$

$$(a_{21} u'_1 + a_{22} u'_2) \cdot u_1 = u_2 \cdot u_1 = 0 \Rightarrow a_{22} = \pm 1$$

$$a_{21} u'_1 \cdot u_1 + a_{22} u'_2 \cdot u_1$$

$$\|u_1 = \pm u'_1\|$$

$$\pm a_{21} u'_1 \cdot u'_1 \pm a_{22} u'_2 \cdot u'_1 = \pm a_{21}$$

Observation (2) (Least squares via QR)

A: linearly
ind.
columns

$$A = QR$$

$$A^T = R^T Q^T$$

$$A^T A x = A^T v$$

$\Downarrow$

$$R^T Q^T Q R x = R^T Q^T v$$

$\Downarrow (R^T)^{-1}$

$$Q^T Q R x = Q^T v$$

$\Downarrow$

$$Q^T Q = I_m \text{ (orthonormality)}$$

$$R x = Q^T v$$

This always has a solution because

R is invertible

Cauchy-Schwarz inequality

Theorem

$$u, v \in \mathbb{R}^n$$

$$|u \cdot v| \leq \|u\| \cdot \|v\|$$

Req: Isn't $u \cdot v = \|u\| \cdot \|v\| \cos \theta$?

$$|\cos \theta| \leq 1 \text{ !?}$$

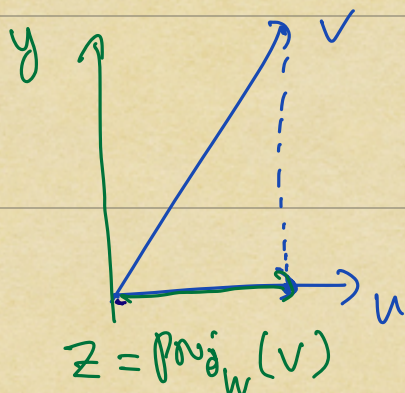
This is actually justified by the
Cauchy-Schwarz inequality

Pf:

If $u=0$ or $v=0$, then the
inequality is trivial

$\Rightarrow$ Can assume both u & v are non-zero

$$\|u\| \geq 0$$



$$W = \text{Span}(u) \leq \mathbb{R}^n$$

Recall: If W has an orthonormal basis $u_1, \dots, u_r$

then $\text{proj}_W(v) = (v \cdot u_1)u_1 + \dots + (v \cdot u_r)u_r$

If $W = \text{span}(u)$, orthonormal basis is $\left\{ \frac{u}{\|u\|} \right\}$

$$\Rightarrow \text{proj}_W(v) = \left(v \cdot \frac{u}{\|u\|} \right) \cdot \frac{u}{\|u\|}$$

$$= \frac{(v \cdot u) \cdot u}{\|u\|^2}$$

$$\left[= \frac{(v \cdot u)}{(u \cdot u)} u \right]$$

$$y = v - \text{proj}_W(v) = v - \frac{(v \cdot u)}{\|u\|^2} u$$

$$y \cdot y = (v - \text{proj}_W(v)) \cdot (v - \text{proj}_W(v))$$

$$= v \cdot v - \underbrace{\text{proj}_W(v) \cdot v - \text{proj}_W(v) \cdot v}_{=0} + \text{proj}_W(v) \cdot \text{proj}_W(v)$$

$$\text{proj}_W(v) \cdot v = \frac{(v \cdot u)}{\|u\|^2} \cdot (v \cdot u) = \frac{|(v \cdot u)|^2}{\|u\|^2}$$

$$\text{proj}_W(v) \cdot \text{proj}_W(v) = \left(\frac{(v \cdot u)}{\|u\|} \cdot u \right) \cdot \left(\frac{(v \cdot u)}{\|u\|} \cdot u \right)$$

$$= \frac{(v \cdot u)^2}{\|u\|^2}$$

$$0 \leq \|y\|^2 = y \cdot y = \|v\|^2 - 2 \frac{(v \cdot u)^2}{\|u\|^2} + \frac{(v \cdot u)^2}{\|u\|^2}$$

$$\Rightarrow \frac{1}{\|u\|^2} \quad 0 \leq \|u\|^2 \cdot \|v\|^2 - 2(v \cdot u)^2 + (v \cdot u)^2$$

$$\Rightarrow (v \cdot u)^2 \leq \|u\|^2 \cdot \|v\|^2$$

same
roots

$$|u \cdot v| \leq \|u\| \cdot \|v\|$$

Proposition (Triangle inequality)

$$u, v \in \mathbb{R}^n$$

$$\|u + v\| \leq \|u\| + \|v\|$$

This is a consequence of the Cauchy-Schwarz inequality.